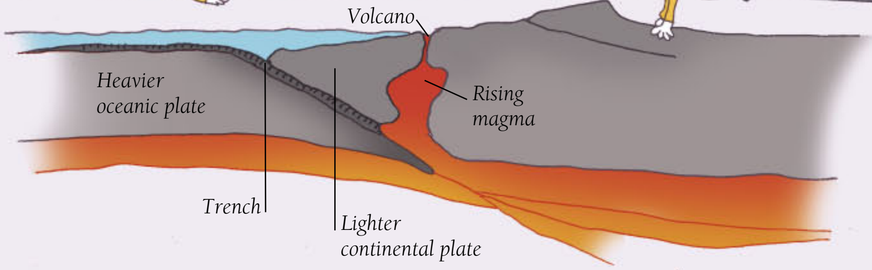
Trenches

The deepest points on Earth, ocean trenches form when a heavier oceanic plate subducts or dips beneath a lighter continental plate.



Volcanoes

When an oceanic plate slides under a continental plate, it melts in the intense heat and forms hot, molten magma that is forced up to the surface and erupts in a violent display to create a volcano.

Ocean waves

As the wind blows across the sea, it causes ripples on the surface. If the wind is strong enough and blows far enough across the water, these ripples build up into waves.

Seawater

Making up 3.5 percent of seawater are dissolved mineral salts—mainly sodium chloride (known to us as table salt) with small amounts of magnesium, sulfur, calcium, potassium, and other elements.

Hot-spot volcano

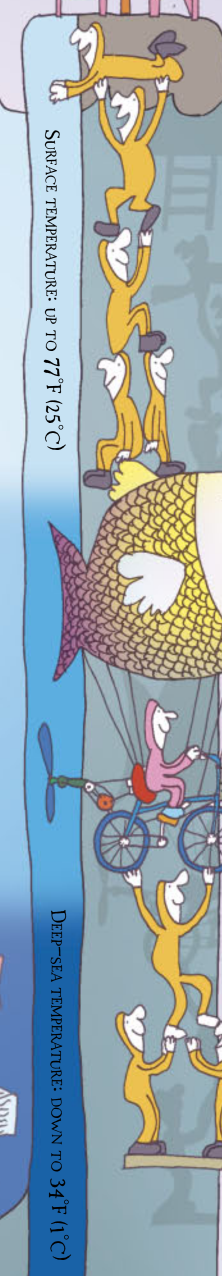
Hot-spot volcanoes form where a stationary hot spot, or hot area in Earth's mantle—the molten rock that lies beneath Earth's crust—burns a hole in the moving tectonic plate above it, creating a line of volcanoes.

Deep-sea exploration

In 1930, the first submersible—the circular bathysphere—was invented, providing the means for exploring the ocean floor. In 1960, the *Trieste* bathyscaphe dived to a record-breaking 35,798ft (10,911m) in the Mariana Trench in the Pacific Ocean.

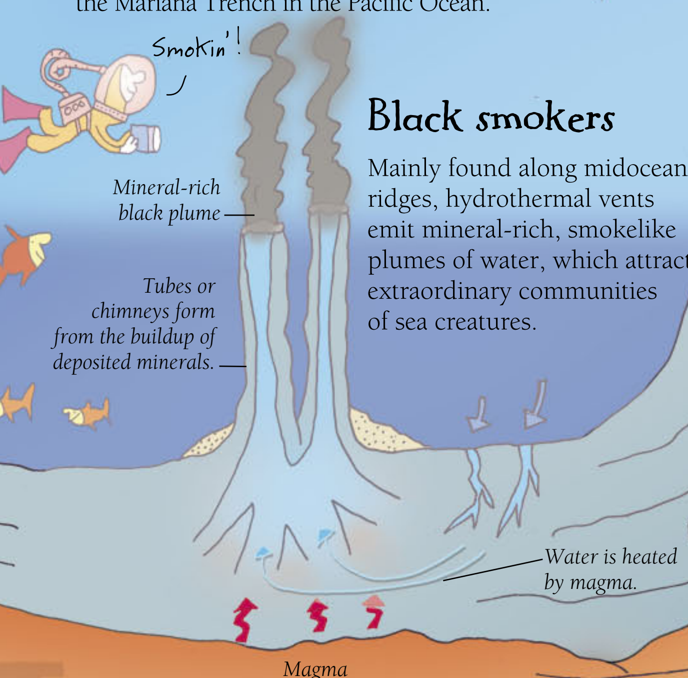
Temperature

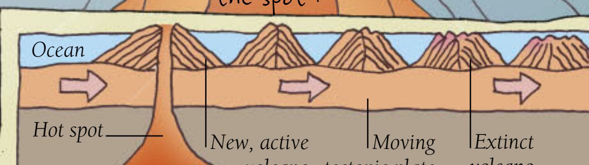
The Sun's warmth can penetrate only so far beneath the ocean waves, and so the deeper you sink, the colder it gets.



Black smokers

Mainly found along midocean ridges, hydrothermal vents emit mineral-rich, smokelike plumes of water, which attract extraordinary communities of sea creatures.





Moving volcanoes

A "hot spot" can burn a hole in a moving tectonic plate to form a volcano. As the plate continues to move, the volcano moves away from the hot spot and becomes extinct. The hot spot then creates a new volcano and the process continues, resulting in a string of extinct volcanoes.

Would you like some mineral water?

Pass the salt.

That's hot!

I've overheated.

I've discovered something.

This is the spot!

We're on the move.